

An Introduction to Causal Inference

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1 Structural Causal Models

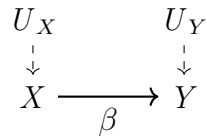
A relationship

$$y = \beta x + u_Y$$

is insufficient to infer any kind of causal relationship between x and y , because rearranging the terms gives

$$x = \frac{y - u_Y}{\beta}$$

To rectify this, a graph can be drawn which encodes any assumptions made about causes and their perceived effects.



U_X and U_Y are called **exogenous** variables, which are those factors that affect variables of interest, but whose influence cannot be measured/explained. They are not the same thing as regression residues, which are independent statistical artifacts. This diagram encodes the facts that

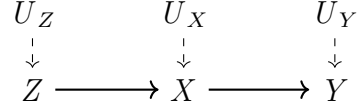
1. X is influenced by U_X
2. Y is influenced by U_Y
3. Y is not influenced by U_X
4. X is not influenced by U_Y
5. U_X and U_Y do not influence each other
6. A unit increase in X , given a fixed value of U_Y , will increase the value of Y by β

A set of nodes S is said to **block** a path p if

- Any arrow-emitting node in p is contained in S

- No member of S is a the direct descendent of a collider (node with 2 parents on the path)

Consider the diagram



If we consider the path $U_Z \leftarrow Z \leftarrow X \leftarrow Y$, it is blocked by both $\{Z\}$ and $\{Y\}$. It is said that U_Z and Y are *d-separated* by S , and which is denoted by $U_Z \perp\!\!\!\perp Y|Z$. This can be read as ' U_Z does not provide any additional information about Y given Z '.

The path $U_Z \leftarrow Z \leftarrow X \rightarrow U_X$ is blocked by ϕ , but not blocked by $\{Y\}$ because it is a descendent of the collider X . This is called Berkson's paradox, where observation of a common consequence makes two independent causes dependent. Flipping either of a pair of fair coins is an independent event, but not with the knowledge that they resulted in at least one tail.

Most generally, the above model can be written using the following equations

$$z = f_Z(u_Z) \tag{1}$$

$$x = f_X(u_X, z) \tag{2}$$

$$y = f_Y(u_Y, x) \tag{3}$$

Joint independence is the only assumption made on u_X, u_Y, u_Z . The absence of a variable as an argument on the right hand side implies that any changes to that variable do not influence the left hand side, assuming that the quantities present on the right hand side are the same.

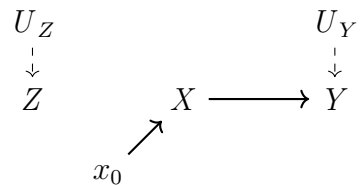
2 The *do* Operator

The *do* operator represents a physical intervention which fixes a particular variable to have a certain value. In a trial for a new drug, $do(X = 1)$ might be used to represent the experiment group. The effect $do(X = x_0)$ has on the model above would be

$$z = f_Z(u_Z) \tag{4}$$

$$x = x_0 \tag{5}$$

$$y = f_Y(u_Y, x) \tag{6}$$



For a model M with a joint distribution $P_M(x, y, z)$, the *do* operator would have the following effect. $P_M(y, z|do(x = x_0)) = P_{M_{x_0}}(y, z)$

A quantity $Q(X)$ is called **identifiable** for a set of assumptions A if

$$P_{M_1}(X) = P_{M_2}(X) \implies Q_{M_1}(X) = Q_{M_2}(X)$$

The model after intervention gives us

$$\mathbb{E}[Y|do(x_0)] = \mathbb{E}[f_Y(u_Y, x_0)]$$

. Checking against the original model

$$\mathbb{E}[Y|X = x_0] = \mathbb{E}[f_Y(u_Y, x)|X = x_0] = \mathbb{E}[f_Y(u_Y, x_0)|X = x_0] = \mathbb{E}[f_Y(u_Y, x_0)] \quad (7)$$

In this case, the expectation is an identifiable quantity, given the assumptions that U_Z is independent of U_X and U_Y , and that Y is a function of X only. It is trivial to check that this would hold for any other model with these assumptions.

3 The Markov Condition

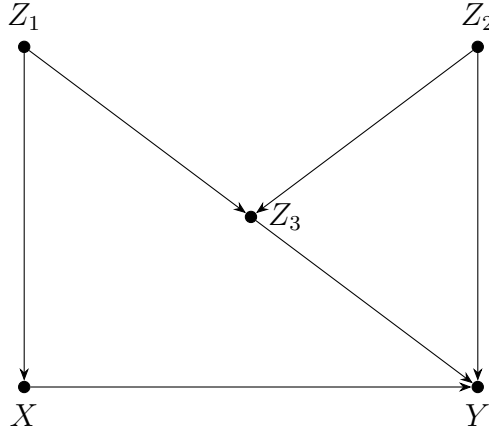
Any distribution generated by a markovian SCM can be decomposed as

$$P(v_1 \dots v_n) = \prod_i P(v_{i+1}|par v_i)$$

where *par* represents the (endogenous) parents of node v_i . More importantly, when making an intervention with the *do* operator, this changes as

$$P(v_1 \dots v_n|do(x = x_0)) = \prod_{i|v_i \notin X} P(v_{i+1}|par v_i)|_{x=x_0}$$

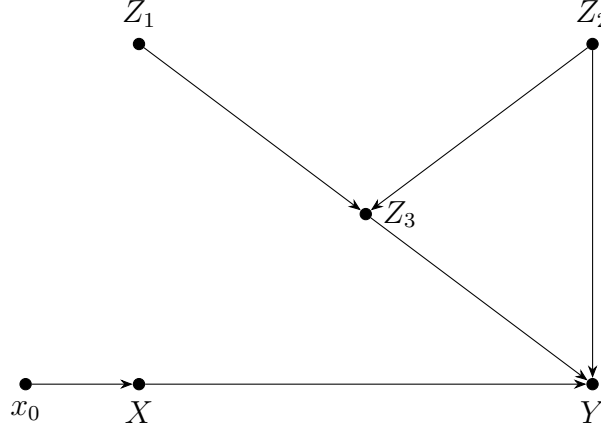
This means that when conditioning on X_0 , all nodes in X must be removed, and all the remaining terms in the product must be evaluated at the point representing the intervention.



Factorising the above model, we get

$$P(x, z_1, z_2, z_3, y) = P(z_1)P(z_2)P(z_3 \mid z_1, z_2)P(x \mid z_1, z_3)P(y \mid z_2, z_3, x) \quad (8)$$

Intervening with $doX = x_0$ changes the diagram as

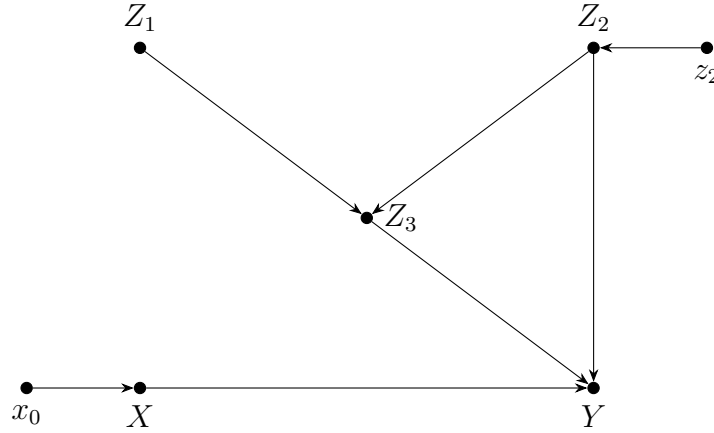


$$P(z_1, z_2, z_3, y | do(x)) = P(z_1)P(z_2)P(z_3 \mid z_1, z_2)P(y \mid z_2, z_3, x_0) \quad (9)$$

Marginalising over the Z values would give us the causal effect of x on y .

$$P(y | do(x)) = \sum_{z_1, z_2, z_3} P(z_1)P(z_2)P(z_3 \mid z_1, z_2)P(y \mid z_2, z_3, x_0) \quad (10)$$

Interventions need not be restricted to a single variable. In the same example, let us intervene with both X and Z_2 .



The distribution post-intervention

$$P(z_1 z_3, y | do(x), do(Z_2 = z)) = P(z_1)P(z_3 \mid z_1, z)P(y \mid z, z_3, x_0) \quad (11)$$

and the causal effect

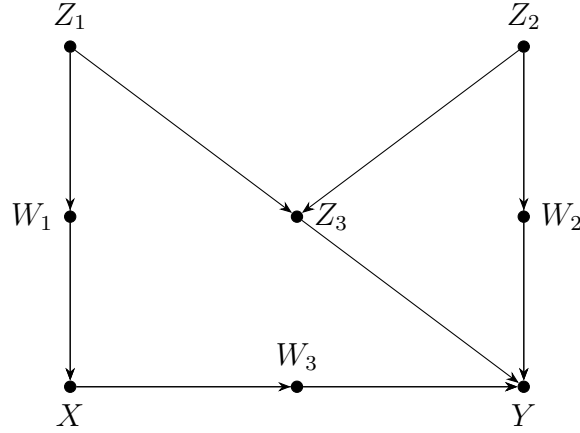
$$P(y | do(x), do(Z_2 = z)) = \sum_{z_1, z_3} P(z_1)P(z_3 \mid z_1, z)P(y \mid z, z_3, x_0) \quad (12)$$

4 Backdoors and Dealing with unmeasurables

In practice, measuring the effect of every relevant counfounder is unfeasible. When trying to measure the causal effect of a variable X on another variable Y with some combination of measurable and unmeasurable influences, we find that there are certain criteria when choosing a set of variables S for adjustment.

1. Nothing in S descends from X .
2. The elements of S block all paths p that end in X

The paths in the second point are called back-door paths, and these are called the **backdoor criteria**.



In this diagram $\{Z_3\}$ blocks $X \rightarrow W_1 \rightarrow Z_1 \leftarrow Z_3 \leftarrow Y$, but not $X \rightarrow W_1 \rightarrow Z_1 \rightarrow Z_2 \leftarrow W_2 \leftarrow Y$. The sufficient sets for X on Y are $\{Z_1, Z_2, Z_3\}$, $\{Z_1, Z_3\}$, $\{W_1, Z_3\}$, $\{W_2, Z_3\}$. Intuitively, stratifying on S removes all bias in measuring the effect of X on Y .

If W_3 is the only obeservable in the above diagram, there is no sufficient adjustment set. However, by first measuring $P(w_3|do(x))$ and then measuring $P(y|do(w_3))$, the backdoor criterion ensures that the relationship is purely causative. This is because every backdoor path from X to W_3 is blocked, and X is a sufficient set for the effect of Y on X .

$$P(y|do(x)) = \sum P(y|do(w_3))p(w_3|do(x))$$

An important result is that a sufficient condition for the relationship between X and Y being identifiable ($P(y|do(x))$) is that every path between X and any of its children is in the sufficient set S .

References

- [1] Judea Pearl. Causal inference in statistics: An overview. *Statistics Surveys*, 3:96–146, 2009.